TIPLE CHOICES QUE

Q4	Q5	Q6	Q7
Q13	Q14	Q15	Q16
			Q17
			Q18

Part 2: Essay Questions (20 marks)

ESTION #2

Write instructions to test bit-5 in register AL, if it is one reset bit-1 otherwise set bit-2

Write a procedure to store '*' into 1 KB memory locations defined as below:

DB 1024 DUP(?)

Using BIOS interrupt, write assembler codes to clear screen and set cursor to position (10, 20).

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رقم الطالب _____ اسم الطالب: _____

1. To transfer value (6F17) 16-BIT data from microprocessor to output port with address 3F0H the following instructions can be used:

A. MOV DX, 03F0H; MOV BX, 6F17H; OUT DX, BX
B. MOV DX, 03F0H; MOV AX, 6F17H; OUT DX, AX
C. MOV DX, 03F0H; MOV AX, 6F17H; OUT DX, AL
D. MOV DX, 03FCH; OUT DX, AL

2. The interrupt generated by the execution of an interrupt instruction INT is

A. NMI interrupt B. software interrupts C. external interrupt D. Interrupt-in-interrupt

3. The physical address of interrupt vector associated with INT 16 is:

A. 00042H B. 00040H C. 00064H D. 0003CH

4. The size of interrupt vectors table is ------

A. 4 bytes B. 1024 bytes C. 64 bytes D. 128 bytes

5. The binary value of AL and carry flag (CF)after the following instructions have executed :

```
MOV AL,0ACH
SAR AL,1
```

A. AL=01010101B; CF=1
B. AL=11010110B;CF=0
C. AL=01010100B; CF=0
D. AL=11010110B;CF=1

6. The binary value of AL and CF after the following instructions have executed is _____

```
MOV CL,4
MOV AL, 7FH
ROL AL,CL
```

A. AL=11110111B; CF=0
B. AL=11110111B; CF=1
C. AL=01111111B;CF=1
D. AL=01111111B CF=0

7. In the following instruction sequence, show the value of AL, in binary:

```
MOV AL,3FH
NEG AL
XOR AL,FFH
```

A. AL= 00111110B B. AL= 01011111B C. AL= 01011110B D. AL= 01011111B

8. If there is a carry from lowest 4 bit during addition, _____ flag sets.

A. Auxiliary carry B. CarryC. Over flow D. Sign

9. The result of multiplication instruction MUL BX is stored in register pair

A. DX:AX B. AX:DX C. CX:AX D. AX:CX

10. Find the content of register AL after execution of the following code:

```
MOV AL,11
SHR AL,1
ADC AL,1
```

A. 7 B. 5 C. 4 D. 3

11. Find the content of memory location X after execution of the following code

```
X DB ?
MOV AX,11
MOV BL,3
DIV BL
OR AH,30H
MOV X,AH
```

A. X='2' B. X=0 C. X= '0' D. X=2

12. Find the content of AH registers after execution of the following code:

```
MOV AX,14
MOV BL,4
DIV BL
XOR AH,0FH
```

A. D0H B. C0H C. 0DH D. 0CH

13. LOPPE instruction repeat a group of instruction while

A. (CX=0)&(ZF=0) B. (CX=0) & (ZF≠0) C. (CX≠0) &(ZF=1) D. (Cx≠0) & (ZF=0)

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14. The value of CX and AL after executing the following code is:

```
        MOV AL, FFH
        MOV CX, 6
L1:     ADD AL, 1
        LOOPZ  L1
```

A. CX=4; Al=2 B. CX=2; Al=3 C. CX=5; Al=1 C. D. CX=3; AL=6

15. To compare two strings in equality we use prefix ----------before instruction CMPSB:

A. REP B. REPNE C. REPE D.REPNZ

16. The equivalent instructions of STOSB; assuming DF flag =0:

A. MOV AL,[SI]; INC SI B. MOV AL,[SI]; DEC SI

C. MOV [DI],AL; INC DI D. MOV [DI],AL; DEC DI

17. What will be appears on the screen after executing the following code:

```
STR DB "CDEFGHK", 0DH, "AB"
    DB 'XYZ' , '$'
MOV AH, 9
MOV DX, OFFSET STR
INT 21H
```

A) CDEFGHAB B) ABXYZH C) ABCDEFGH D) ABXYZHK

18. The values of array Y after executing the following code is:

```
X DB   'ABC+CDK'
Y DB    5 DUP('*')
CLD
MOV SI, OFFSET X
MOV DI, OFFSET Y
MOV CX, 3
REP MOVSB
MOV AL, '#'
STOSB
```

A) Y=ABC+CDK B) Y=ABC*# C) Y=ABC#*CDK D) Y=ABC#*

19. The value of Y after executing the following code is:

```
X DB  4,7,8
Y   DB  ?
MOV  BX,OFFSET  X
MOV CX, 2
MOV  AL,0
L:  CALL XXX
    LOOP L
    MOV Y, AL
    HLT
    XXX   PROC
                MOV AH, [BX]
                INC BX
                ADD AL, AH
                RET
    XXX  ENDP
```

A) 11 B) 04h C) 0 D) 9

20. In asynchronous serial data transmission, assume The data format is: 8 data bit, 1 start bit, 1 stop bit and even parity, What is the speed of transmission (baud rate) if the device transmit 20 characters per second?

A) 220 B) 20 C) 180 D)200

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- Write an assembler code to reverse string X in order; use LODSB and STOSB strings; where
  X DB 'ABCDEK+++*'

QUESTION#3

Interface 8- bit input port (74LS245) to read the status of switches SW1 to SW8 to the microprocessor system. Store the status in memory location X defined as byte. Assume the address of the port is 08FFH.

a) Draw block circuit diagram circuit, showing address decoding logic, and control signals
b) Write a program to read the status of switches and store the result into X memory location

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